Reading report

PhD candidate: Maxime Deforche
Title of the doctoral dissertation: Management and Analysis of Short Texts: A Graph-Based Approach

Member of the examination board: Walter Daelemans

The dissertation is well-written and presents solid and coherent research results. It provides a nice content consisting of complementary theoretical/algorithmic work combined with empirical testing on a difficult corpus of short texts in a less-resourced language, a corpus containing in addition lots of partial overlap and incomplete metadata.

The research ticks all the required boxes for a dissertation: the innovation of the contributions is shown in the context of a (as far as I can judge) a sufficiently complete state of the art analysis, methodology and data are clearly described and are fit for the research goals, and the research results have potentially a high impact in data science and NLP. The availability of data and software is commendable and makes the research replicable.

The organization of the thesis is centered around 5 chapters based on published/submitted papers. I very much appreciate that the build-up is sequential, for which published papers had to be redistributed over the chapters (not simply one paper = one chapter). This adds a lot to comprehensibility. Likewise, the introductory and concluding chapters have a clear function in the whole, they are not mere appendices, and add context, insight and reflection.

The research objectives and goals of the dissertation are formulated clearly and are important. However, they also were clearly guided by the specifics of this case study (the Database of Byzantine Book Epigrams). That is fine, but it would have been useful if the developed methods were also evaluated on a few other case studies to show generalization. While this limitation is acknowledged and referred to future research, *several other text types are mentioned as potentially useful (social media posts, product reviews, song lyrics, news headlines, and emails), for which many corpora are easily available, so an evaluation or even only a sanity check of the developed methods on these text types would have been possible*. It would also be useful to have some measurement of feasibility of the approach to these corpora based on your computational complexity analyses.

The problem of finding overlapping meaning in different text (1.3.3) is actually a huge topic in literary science and consequently in digital humanities (the problem of intertextuality). This could be added as a potential additional application of the approach.

Regarding research methods, I think it was *a missed chance not to compare with (contextual) text embeddings, the standard approach in NLP for text similarity*. Embeddings are often referred to in the thesis but not head-on compared to the developed methods. Even on small available data, they could be competitive, and they are also hierarchical (implicitly they can be used to represent all the hierarchical levels), and they can also be trained with contextual information and are to a certain extent interpretable. I'm not an expert on Greek data, but sources like Open Greek and Latin, Perseus, ancient Greek treebanks (with universal dependency annotations), Proiel treebank, etc. seem at least partially to contradict your reservations about their possible applicability.

LPG, OWA, L-grades, and how they combine similarity and confidence to handle missing and uncertain data responsibly, are important innovations that make this thesis an important addition to the state of the art.

As mentioned earlier, I see the lack of testing generalization of the approaches to other small text corpora, and the lack of comparison to other baselines, such as embeddings, as the main shortcomings, but don't see a reason to require a new version of the thesis with their addition. I'm fine with the dissertation as it is.

Some additional random remarks / questions

- I was missing a discussion of register and genre of texts as an important source of variability.
- I would have preferred a single bibliography covering all chapters rather than one per chapter.
- You mentioned Pagerank possibilities in your approach, but you don't seem to have used it in this case study, could it have been useful for some research questions?